Practice Exam Version #1

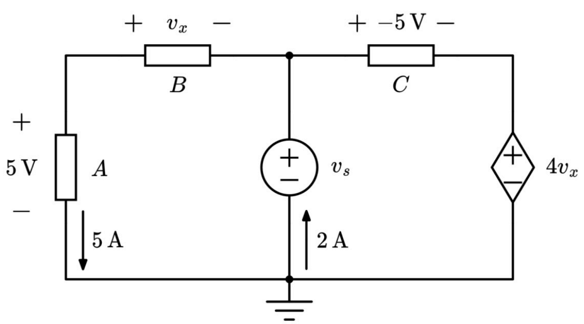
Question One: KCL, KVL, Ohm's Law, and Passive Sign Convention

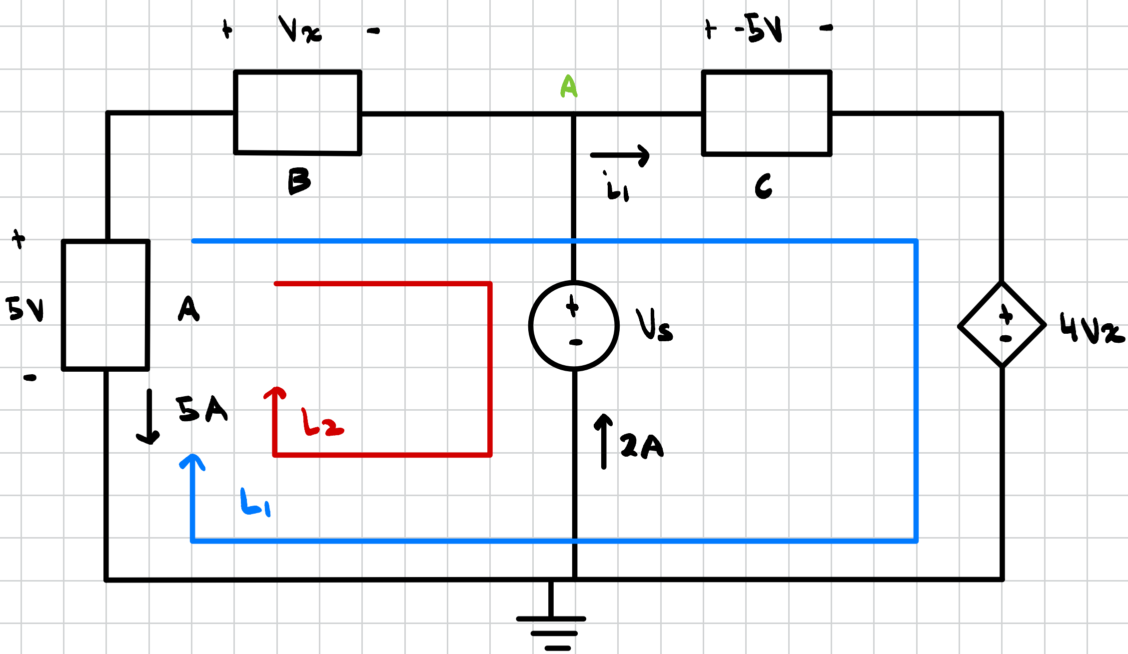
For the circuit shown below, with three unknown circuit elements, find the following:

a) v_x
b) v_s
c) The resistance of circuit element A

Finally, calculate the power of the voltage-dependent voltage source and state whether the power is absorbed or delivered.

Suggested time limit: 20 minutes





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Step #1 KVL @ L_1

$-5 + V_x + (-5) + 4V_x = 0$

$-10 + 5V_x = 0$

$5V_x = 10$

$V_x = 2V$

Step #2 KVL @ L_2

$-5 + V_x + V_s = 0$

$-5 + 2 + V_s = 0$

$-3 + V_s = 0$

$V_s = 3V$

Step #3 R @ Circuit Element A

$V = IR$ where $V = 5V$ and $I = 5A$

$R = \frac{V}{I}$

$R = \frac{5}{5}$

$R = 1\Omega$

Step #4 KCL @ Node A

$5 - 2 + i_1 = 0$

$3 + i_1 = 0$

$i_1 = -3A$

Step #5 Power of Dependent Source

$P = VI$ where $V = 4V_x$ and $I = i_1$

$P = (4)(2)(-3)$

$P = -24W$ Delivered

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Grading Rubric

Find V_x - 3 Points
Find V_s - 3 Points
Find R For Circuit Element A - 2 Points
Find Power of Dependent Source - 1 Point
State Delivered Or Absorb - 1 Point

Discussion #1 Problems To Practice

Problem #5
Problem #8
Problem #9
Problem #10
Problem #12
Problem #13

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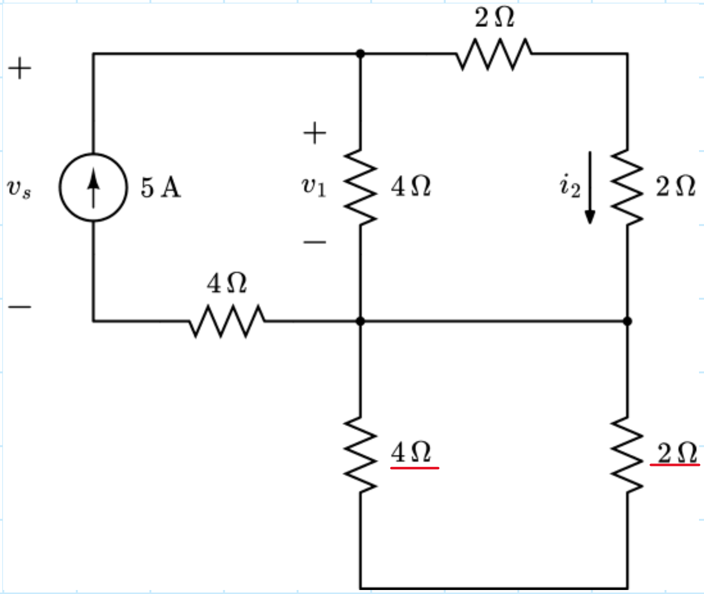
Question Two: Current Division, Voltage Division, and Equivalent Resistance

For the circuit shown below, find the following:

a) v_s

b) v_1

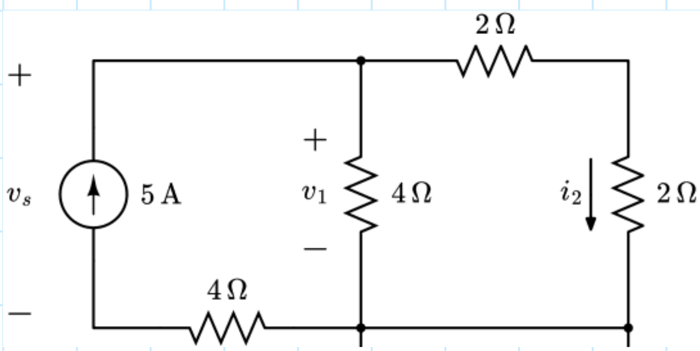
c) i_2



The branch containing the 4Ω and 2Ω resistor is shorted.

If you combined the 4Ω and 2Ω resistor in series, the resultant 6Ω resistor would be connected to the same node on both sides.

Step One: Find the equivalent resistance seen by the independent source



I. $2\Omega + 2\Omega = 4\Omega$

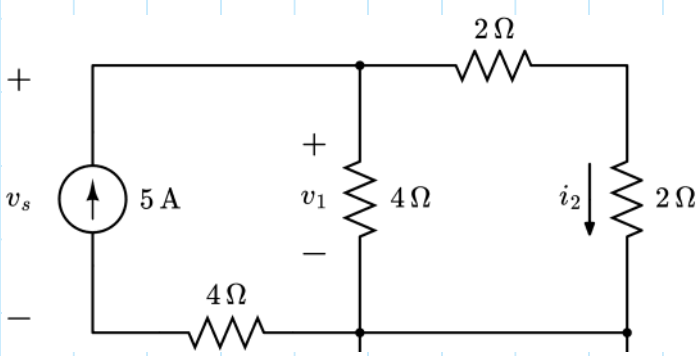
II. $4\Omega \parallel 4\Omega = 2\Omega$

III. $2\Omega + 4\Omega = 6\Omega$

IV. Ohm's Law: $V_s = 5A \cdot 6\Omega$

$V_s = 30V$

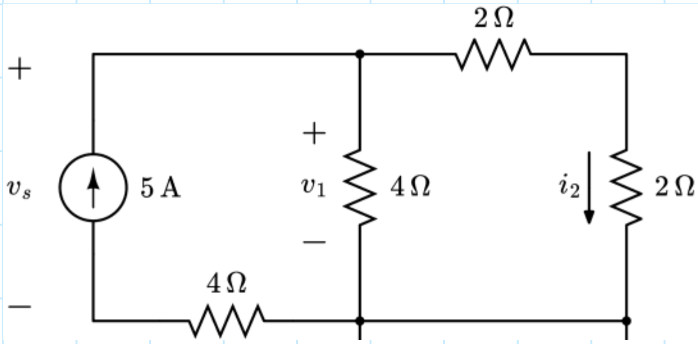
Step Two: Current Division to find i_2

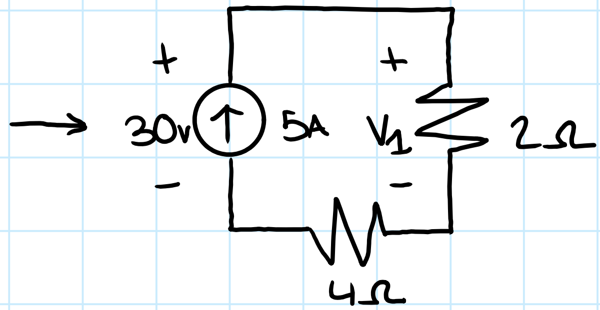


$$i_2 = 5A \cdot \frac{2\Omega}{4\Omega}$$

$$i_2 = 2.5A$$

Step Three: Find V_1 using simplification and voltage division





$$V_1 = 30v \cdot \frac{2\Omega}{6\Omega} = 10v$$

Self-Grading Rubric	
Identify the two shorted resistors	+1 point
Correct equivalent resistance seen by source	+3 points
Identify v_s = 30 V	+2 points
Perform current division to find i_2 = 2.5 A	+2 points
Perform voltage division to find v_1 = 10 V	+2 point

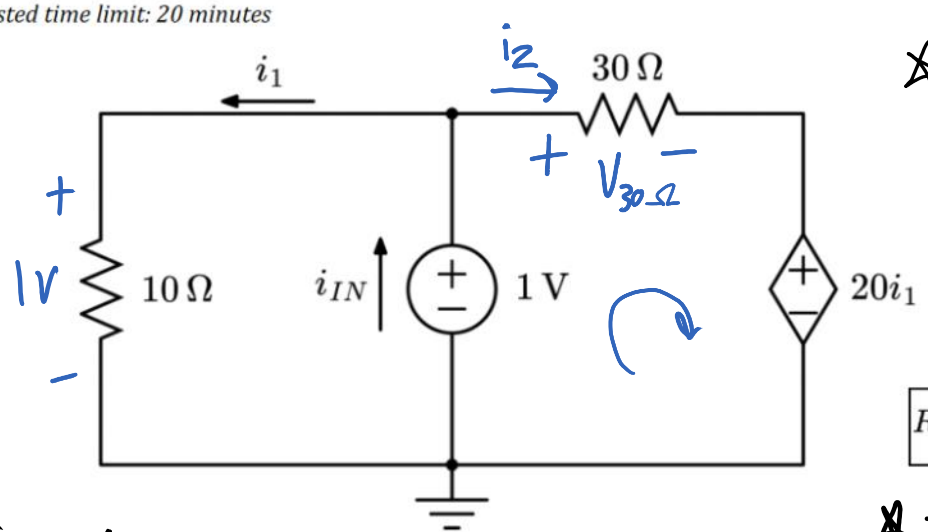
Suggested discussion problems:
- Discussion 2 Questions 7–8
- Discussion 3 Questions 8–10

Practice Exam 1 Version One Q3 Solution

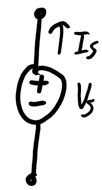
Question Three: Equivalent Resistance with Dependent Sources

For the circuit shown below, find the equivalent resistance seen by the independent voltage source.

Suggested time limit: 20 minutes



★ To find R_{eq} with a dependent source:



$$R_{eq} = \frac{V_s}{I_s}.$$

$$\boxed{R_{eq} = \frac{v_{IN}}{i_{IN}}}$$

Finding i_1:

The 10Ω resistor is parallel with the 1V source, so we know it shares the same voltage!

$$i_1 = \frac{V}{R} = \frac{1}{10} = \underline{0.1\,A}$$

Let's define a current i_2 and a voltage $V_{30\Omega}$.

KVL right loop:

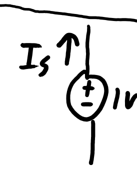
$$-1 + V_{30\Omega} + 20\overset{0.1\,A}{i_1} = 0$$

$$-1 + V_{30\Omega} + 2 = 0$$

$$\underline{V_{30\Omega} = -1V}$$

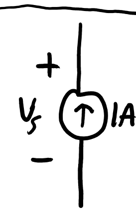
★ If we give V_s or I_s an arbitrary value (such as 1V or 1A), we can solve for the other value to get R_{eq}.

For 1V source,



$$R_{eq} = \frac{1}{I_s}$$

For 1A source,



$$R_{eq} = \frac{1}{V_s} = V_s.$$

★ You can choose any way you want! A 1V source is given here, so let's use that.

- Note that I_s is the same as i_{IN} here.

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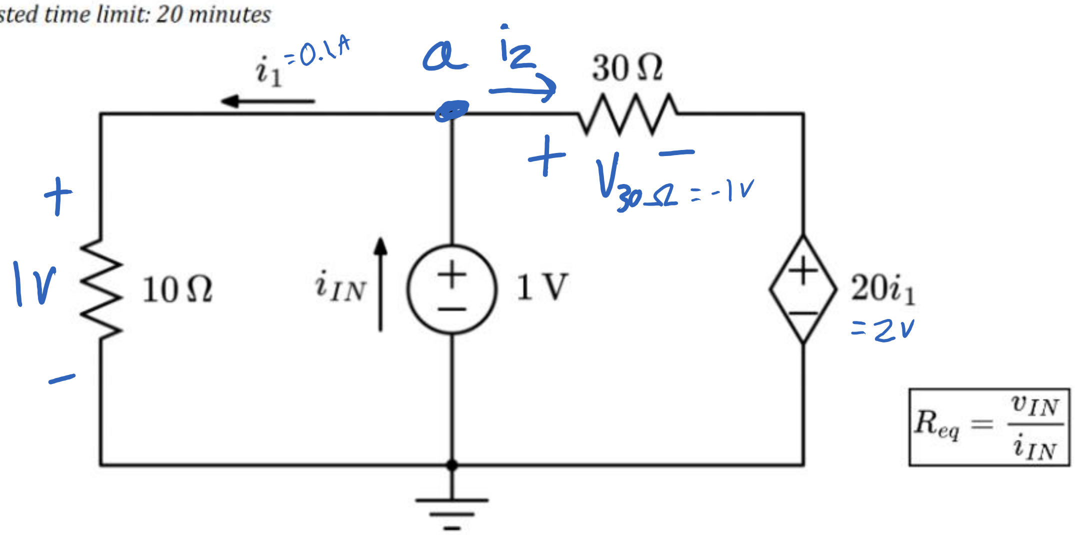
Question Three: Equivalent Resistance with Dependent Sources

For the circuit shown below, find the equivalent resistance seen by the independent voltage source.

Suggested time limit: 20 minutes



$$i_2 = \frac{V_{30\Omega}}{R} = \frac{-1}{30} A.$$

KCL @ a:

$$i_1 + i_2 - i_{IN} = 0$$

$$0.1 + \left(-\frac{1}{30}\right) - i_{IN} = 0$$

$$\boxed{i_{IN} = \frac{1}{15} A.}$$

$$R_{eq} = \frac{1}{I_s} = \frac{1}{i_{IN}} = \frac{1}{\frac{1}{15}} = 15\Omega.$$

$$\boxed{R_{eq} = 15\Omega.}$$

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Grading Rubric

Find i_1: +2 Points

Find $V_{30\Omega}$: +2 Points

Find i_2: +1 Point

Find i_{IN}: +2 Points

Find R_{eq}: +2 Points

Discussion 3 Problems To Practice:

- Problem 4
- Problem 5
- Problem 6
- Problem 7

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